פ0 — From the Boundary: The Place-Completion of the Apex

Leonardo Murillo Montero*ITCR, UNA, UCR*

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Abstract

Situation. Six exterior papers read the categorical apex through its boundary, one local arithmetic at a time: the conductor packets and units (פ3, פ4), the Gauss sums at scale $\sqrt{N}$ (פ4), the radical–cyclotomic seam (פ5), the archimedean wall — the smooth primorial and its moment barrier (פ3, פ6). Each is a *local place* of $\mathbb{Q}$. In parallel, the interior's Paper 27 assembles a conditional Riemann-Hypothesis reduction on the primorial tower: a canonical involution implementing $s \leftrightarrow 1-s$, recovery of the Weil explicit-formula prime sum (Thm 3.1, under hypotheses PC1–PC4), and an archimedean completion isolated in Axioms A1–A5.

Gap. The six papers study *discs* — one place, fixed N. Neither they nor Paper 27 assemble the **single object over all places** of which the local boundaries are the factors. That object is the adele $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p$, self-dual under Fourier, from which the completed $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its functional equation descend (Tate). This keystone — written after פ1–פ6 as ל0 was written after ל1–ל6 — assembles it.

Action. Under the grounded-paranoid-delusional method (Murillo Montero 2026): a fenced *Star* (§2); four matched-pair falsifiers (§3); one deterministic exact artifact (`P0-settlements.py`) catching the *unconditional* shadows of Paper 27's reduction (§4). The classical global theorems (Tate, Weil) are **cited, not claimed**; RH is **named, not claimed**.

Result. The **Place-Completion Principle**: each local boundary of פ1–פ6 is the local factor of one self-dual adelic object, whose self-duality *is* the functional equation; the exterior ∂, globalized, is the diagonal $\mathbb{Q} \hookrightarrow \mathbb{A}$. Four exact catches anchor it. **פ0-involution (K).** Paper 27's canonical involution ι (frequency reversal $a \mapsto -a$) is exactly F^2, the *square of the finite Fourier transform* ($F^4 = I$); its $+1$-eigenspace — the finite critical-line sector — has dimension $\lfloor N/2 \rfloor + 1$. **פ0-rootnumber (K, joint; naive falsified).** $\mathrm{tr}(F) = g/\sqrt{N}$ is the quadratic Gauss sum; for odd N, $|g|^2 = N$ — פ4's $\sqrt{N}$ root-number scale — but the naive "always $\sqrt{N}$" is *false*: $g = 0$ for $N \equiv 2\ (4)$. **פ0-positivity (K).** F is unitary ($F^*F = I$ exactly), so the finite recovered energy is a norm, ≥ 0 for free; Paper 27's "joint" positivity (A5) sharpens to: RH is positivity *surviving the projective limit* once the archimedean Γ-kernel is adjoined — the kernel פ6 proved is not finitely generable. **פ0-functionfield (K).** The Fermat cubic obeys $a^2 \leq 4q$ (Weil) for all tested q: where there is no archimedean place, Paper 27's A1–A5 are all *theorems* and RH holds.

Harvest. The keystone gives the exterior series its completion as T0 gave the interior its coordinate: the local boundaries are one self-dual object; its functional equation is Tate's self-duality; and RH is the positivity of that object once the one archimedean term — the terminal boundary of פ3/פ6 — is adjoined. That the gap is *exactly* that one term is shown by פ0-functionfield: with ∞ removed, the whole reduction closes (Weil). The four catches and the function-field validation are [P]; the place-dictionary is [A]; RH, and Paper 27's hypotheses PC1–PC4 / Axioms A1–A5, stay [C] — named, built toward, not claimed.

Six earlier papers each studied the prime numbers from one angle. This one shows the angles are views of a single object — the way local weather maps assemble into one planet. The object has a perfect mirror symmetry (a known classical theorem); a celebrated unsolved problem, the Riemann Hypothesis, asks whether its hidden resonances all sit on the mirror's axis. We make three things exact and one thing honest. Exact: the "mirror" operation is literally the square of a Fourier transform; the number that measures it is a classical Gauss sum; and the easy half of the hard problem (a positivity) is automatic. Honest: the entire difference between the solved cousin of the problem (over finite systems of arithmetic, settled by André Weil) and the open original is a single ingredient — the "point at infinity" — which an earlier paper proved cannot be built from the primes alone. We do not solve the problem; we locate it, precisely, on that one ingredient. Every number below is the printout of a short exact program.

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Notation

Symbol	Meaning
place v	a prime p (local field $\mathbb{Q}_p$) or ∞ (the field $\mathbb{R}$).
$\mathbb{A}$	adele ring $\mathbb{R} \times \prod_p' \mathbb{Q}_p$; finite part $\varprojlim \mathbb{Z}/P_k = \hat{\mathbb{Z}}$.
$\mathbb{Q} \hookrightarrow \mathbb{A}$	diagonal, discrete, self-annihilating ($\mathbb{Q}^\perp = \mathbb{Q}$) — the global ∂.
$\xi(s)$	completed zeta $\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\xi(s) = \xi(1-s)$ (Tate).
F	finite Fourier transform on $\mathbb{Z}/N$, $F_{jk} = \omega^{jk}/\sqrt{N}$, $\omega = e^{2\pi i/N}$.
ι	the canonical involution $s \leftrightarrow 1-s$; finite form $x \mapsto -x = F^2$ (Paper 27).
g	quadratic Gauss sum $\sum_k \omega^{k^2}$; $\mathrm{tr}(F) = g/\sqrt{N}$.
$Z_v(s)$	local Tate factor: $(1-p^{-s})^{-1}$ at p; $\pi^{-s/2}\Gamma(s/2)$ at ∞.
a, q	Frobenius trace $a = q + 1 - \#C(\mathbb{F}_q)$ of the Fermat cubic; Weil $a^2 \le 4q$.
PC1–4, A1–5	Paper 27's projective-closure hypotheses and Archimedean-limit axioms (conditional).
[P]/[A]/[C]	Tiers: proven / structural-pending-named-audit / conjectural-reduced.
K, L	Matched-pair branches; *joint-on-disjoint* when both fire on disjoint sub-structures.

Acronyms

Short	Expansion
GPD	Grounded-paranoid-delusional (the method of Murillo Montero 2026).
(G)RH	(Generalized) Riemann Hypothesis.
adele / idele	$\mathbb{A} = \prod_v' \mathbb{Q}_v$ and its units — the object holding all places at once.
Tate's thesis	the derivation of $\xi(s) = \xi(1-s)$ from adelic Fourier self-duality.
local factor	the per-place contribution $Z_v(s)$ to $\xi(s) = \prod_v Z_v$.

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1. The reflection is a Fourier square

Begin where Euler would: with the smallest exact fact that already names the whole object.

The functional equation of the zeta function is a *reflection*, $s \leftrightarrow 1-s$. Its finite avatar — the one the interior's Paper 27 builds, with some care, out of Bures geometry — is frequency reversal: send each character χ_a to χ_{-a}, equivalently send a function on the N-hour clock to its mirror $x \mapsto -x$. Compute what operator that is. The finite Fourier transform F has $(F^2)_{jl} = \frac{1}{N}\sum_k \omega^{k(j+l)} = [\, l \equiv -j \,]$ — i.e. F^2 *is* the mirror $x \mapsto -x$. So the reflection is nothing exotic: it is the **square of the Fourier transform**, an order-two element of the order-four cyclic group generated by F ($F^4 = I$). One line replaces a section of construction.

That is the key in miniature. The reflection that the Riemann Hypothesis lives under is the self-duality of the Fourier transform; globally, across all places of $\mathbb{Q}$, it is the self-duality of the adele ring, and the completed zeta function is the object it fixes. The six exterior papers each found one local face of that object — one place. This keystone, written after them as the interior's ל0 was written after ל1–ל6, assembles the faces into the object, names the principle, and locates — precisely, and without claiming — where the Hypothesis sits.

1.1 Why a gate, and why exact

A word on method, sharpened here because the destination is the most over-claimable in mathematics. We permit one unifying picture but give it no vote (§2); we fix the meaning of every outcome before computing (§3); we settle each question with one short exact-arithmetic program — decisions by orthogonality, bijection, and integer point-counting, floating point only as a labelled non-voting witness (§4). Two classical global theorems are used and **cited, not reproved**: Tate's functional equation and Weil's bound for curves. The Riemann Hypothesis is **named and not claimed**, and Paper 27's enabling hypotheses (PC1–4) and axioms (A1–5) are carried at [C], never promoted. The keystone's whole contribution is the honest separation of what is assembled from what remains.

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2. The Star (delusional; no vote on truth)

> **Fence.** This section is *Historic Fiction*. It chooses what to attempt. It decides nothing. The road to a result runs through §3 and §4.

Here is the picture. A number field has two kinds of place: the finite primes p (local fields $\mathbb{Q}_p$) and the real place ∞ ($\mathbb{R}$). The adele $\mathbb{A}$ holds them all at once, as a restricted product; $\mathbb{Q}$ sits inside it diagonally as a

self-annihilating discrete subgroup ($\mathbb{Q}^\perp = \mathbb{Q}$). The exterior boundary ∂ that the whole series chases is, at this altitude, that diagonal — the local–global boundary.

The Place-Completion Principle (the fiction to chase). Each local boundary of ם1–ם6 is the local factor of one global object:

$$\xi(s) = \prod_v Z_v(s),$$

$$Z_p = (1 - p^{-s})^{-1} \quad \text{(the packets/units of ם3, ם4)},$$

$$Z_\infty = \pi^{-s/2}\Gamma(s/2) \quad \text{(the } \log 2\pi \text{ of ם6)}.$$

The single Mellin variable that T0/T23 already isolated as "dual to global scale, shared by all CRT factors, split by none" is the **idele-norm exponent** s; T0 chose a section of it (the gauge β), and this keystone completes it globally. The object is **self-dual**: $\mathbb{A}$ is its own Fourier dual, the standard test state is its own transform, and the completed ξ is *fixed under the reflection* $s \leftrightarrow 1 - s$. That self-duality, integrated against the idele norm (Tate), **is** the functional equation. Its zeros are a discrete spectrum, symmetric about the axis $\mathrm{Re}(s) = \frac{1}{2}$ — the fixed locus of the reflection. The fiction: the Hypothesis is the statement that this spectrum is *real* — that the object is not merely symmetric but self-adjoint, every resonance on the axis.

If true: the six local boundaries are one self-dual whole; its functional equation is the self-duality made an identity (Tate, classical); and RH is the positivity of that whole once the one archimedean factor is folded in. The paper does not argue this. It catches four exact, unconditional shadows of it and reports the arithmetic.

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3. Pre-registration (paranoid; meanings fixed before the check)

Audit	Claim under test	K	L	Joint-on-disjoint
ƀ0-involution	The reflection ι is the Fourier square.	$F^2 =$ parity $x \mapsto -x = \iota$ (orthogonality), $F^4 = I$; $+1$-eigenspace dim $= \lfloor N/2 \rfloor + 1$.	$\iota \neq F^2$ / needs the Bures construction.	—
ƀ0-rootnumber	$\mathrm{tr}(F)$ is the Gauss sum = ƀ4's scale.	odd N: $\lvert g\rvert^2 = N$ (bijection) $\Rightarrow \lvert\mathrm{tr}\, F\rvert = 1$, the $\sqrt{N}$ scale.	naive: $\lvert g\rvert^2 = N$ for *all* N.	mod-4 trichotomy; naive L falsified ($g = 0$ at $N \equiv 2$).
ƀ0-positivity	Finite positivity is free; RH = limit-positivity.	$F^*F = I$ exact $\Rightarrow$ finite energy ≥ 0 automatically.	finite Gram not PSD.	RH = positivity surviving the limit + the archimedean term.
ƀ0-functionfield	With ∞ absent, A1–A5 are theorems.	Fermat cubic $a^2 \leq 4q$ (Weil) for all q; $\iota = T \mapsto 1/(qT)$; A2 empty.	$a^2 > 4q$ some q (impossible).	—

We run the artifact.

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4. Settlement (grounded; the transcript is the record)

4.1 ƀ0-involution — the reflection is F^2

The finite avatar of $s \leftrightarrow 1 - s$ is the mirror $x \mapsto -x$ on $\mathbb{Z}/N$.

Proposition 4.1 (the involution is the Fourier square). *On $\mathbb{Z}/N$, F^2 is the parity permutation $x \mapsto -x$, hence $F^2 = \iota$ and $F^4 = I$. The $+1$-eigenspace of ι — the reflection-symmetric (finite critical-line) sector — has dimension $\lfloor N/2 \rfloor + 1$.*

Proof. $(F^2)_{jl} = \frac{1}{N}\sum_k \omega^{k(j+l)} = [\, l \equiv -j \,]$ by the orthogonality $\sum_k \omega^{kt} = N\,[t \equiv 0]$ (the artifact verifies this integer relation exactly for every t). Thus F^2 permutes $|x\rangle \mapsto |-x\rangle = \iota$, and $F^4 = \iota^2 = I$. The $+1$-eigenspace consists of even functions $f(x) = f(-x)$, of dimension equal to the number of orbits of $x \mapsto -x$, namely $\lfloor N/2 \rfloor + 1$. □

Settlement ƀ0-involution — K, [P]. Exact at $N = 6, 12, 30$ (orthogonality holds; dims $4, 7, 16 = \lfloor N/2 \rfloor + 1$; witnesses $\|F^2 - \text{parity}\|, \|F^4 - I\| \sim 10^{-15}$). Paper 27 builds ι from Bures geometry and verifies four properties; the exterior reading collapses that to one identity — **the canonical involution is the square of the Fourier transform**, unconditionally. The reflection that RH lives under is Fourier self-duality, finitely.

4.2 ƀ0-rootnumber — the Fourier trace is the Gauss sum

The operator whose square is the reflection carries, in its trace, the program's $\sqrt{N}$ scale.

Proposition 4.2 (trace = Gauss sum; the odd-conductor scale). $\mathrm{tr}(F) = \frac{1}{\sqrt{N}}\sum_k \omega^{k^2} = g/\sqrt{N}$. *For odd N, $\lvert g\rvert^2 = N$, so $\lvert\mathrm{tr}\, F\rvert = 1$ — the $\sqrt{N}$ scale of ƀ4's Gauss sums, the local root number. The naive*

extension "$|g|^2 = N$ for all N" is false: $g = 0$ for $N \equiv 2 \pmod 4$ (Gauss's evaluation).

Proof. $\operatorname{tr}(F) = \sum_k F_{kk} = \frac{1}{\sqrt{N}} \sum_k \omega^{k^2}$ by definition. For odd N, 2 is invertible, so $(k, m) \mapsto (k - m, k + m)$ is a bijection of $(\mathbb{Z}/N)^2$ (verified exactly: the image has N^2 elements); since $k^2 - m^2 = (k - m)(k + m)$, $|g|^2 = \sum_{k,m} \omega^{k^2 - m^2} = \sum_{a,b} \omega^{ab} = \sum_a N\,[a \equiv 0] = N$. For $N = 2M$ with M odd, $(k+M)^2 \equiv k^2 + M \pmod N$ (exact integer identity, verified) and $\omega^M = \omega^{N/2} = -1$, so $\omega^{k^2} + \omega^{(k+M)^2} = \omega^{k^2}(1 - 1) = 0$, and pairing $k \leftrightarrow k + N/2$ gives $g = 0$. □

Settlement ℌ0-rootnumber — K + joint-on-disjoint, naive L falsified, [P]. Odd $N = 3, 5, 7, 11, 13$: bijection holds, $|g|^2 = N$, the $\sqrt{N}$ scale of ℌ4. Even N: $g = 0$ at $N \equiv 2$ ($N = 6, 30$), $|g|^2 = 2N$ at $N \equiv 0$ ($N = 8, 12$) — Gauss's mod-4 trichotomy, the honest refinement that kills the naive claim. The reflection operator's trace is exactly the Gauss sum the function-field side (ℌ4) found carrying the critical scale; here it is the local root number of the local functional equation.

4.3 ℌ0-positivity — the finite half is free

Paper 27's A5 asks for *joint* positivity (finite closure and the archimedean involution together). One half is automatic.

Proposition 4.3 (finite positivity is unconditional). *F is unitary, $F^*F = I$ exactly (orthogonality), so every finite recovered energy is a squared norm and is ≥ 0 without hypothesis. Consequently the positivity of the limiting Weil form is not "joint": its finite part is free, and its entire content is whether positivity is preserved in the projective limit once the archimedean factor $1/|\Gamma(\frac{1}{2} + it)|^2$ is adjoined.*

Proof. $(F^*F)_{jl} = \frac{1}{N} \sum_k \omega^{k(l-j)} = [\, l \equiv j \,] = \delta_{jl}$ (verified exactly). A Gram form G^*G is positive semidefinite; the finite Noether/leakage energy of Paper 27 is of this type. □

Settlement ℌ0-positivity — K, [P] finite / [C] limit. $F^*F = I$ exactly at $N = 12, 30$; the finite Gram is the identity, eigenvalues all 1. So RH is the positivity that must *survive the limit* — and the term whose adjunction is in question is exactly the archimedean Γ-factor that ℌ6 proved cannot be generated from the finite primes (the moment barrier). One honest caution, carried forward from ℌ6: this Weil-form positivity is *not* ℌ6's Hankel/moment positivity — they are cousins at the same place, not the same condition, and neither is claimed to settle RH.

4.4 ℌ0-functionfield — the gap is exactly ∞

The cleanest evidence for where RH lives is the case where there is no archimedean place at all.

Proposition 4.4 (with ∞ absent, the reduction closes). *For the Fermat cubic $C : x^3 + y^3 + z^3 = 0$ over $\mathbb{F}_q$, the Frobenius trace $a = q + 1 - \#C(\mathbb{F}_q)$ satisfies $a^2 \leq 4q$ (Weil), so $|\alpha| = \sqrt{q}$ and RH holds. In this setting Paper 27's Axioms become theorems: the involution is the Frobenius functional equation $T \mapsto 1/(qT)$, the infinite-place packet (A2) is empty, and positivity (A5) is automatic.*

Proof. The point count is exact integer arithmetic; $a^2 \leq 4q$ is Weil's theorem for the genus-one curve, verified by counting (the artifact confirms it for $q = 5, \ldots, 43$, with the ordinary/supersingular split). With no place at ∞ there is no Γ-factor to adjoin; the explicit formula is purely finite (Plancherel), positive by §4.3. □

Settlement ℌ0-functionfield — K, [P]. $a^2 \leq 4q$ holds for every tested q. The function-field case is Paper 27's roadmap *unconditionally completed* — A1–A5 all theorems, RH true — and it differs from the number field by exactly one term: the archimedean place, the terminal boundary of ℌ3/ℌ6. **ℌ4 is Paper 27 done, minus ∞.** This is the keystone's sharpest honest statement: the entire distance from proven RH to open RH is the single factor the program has independently proved is not finitely generable.

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5. The synthesis (earned)

Proposition 5.1 (the Place-Completion Principle, at its tiers). *The local boundaries of פ1–פ6 are the local factors Z_v of the self-dual completed object $\xi = \prod_v Z_v$; its self-duality is the functional equation $\xi(s) = \xi(1-s)$ ([P], Tate); the reflection is the Fourier square (§4.1, [P]); its trace is the Gauss-sum root number (§4.2, [P]); the finite positivity is free (§4.3, [P]). That these factors assemble the program's apex into Tate's adelic object is the place-dictionary, [A]. That the spectrum lies on the axis (RH) is [C], with the gap pinned to archimedean positivity (§4.3) and validated, where ∞ is absent, by Weil (§4.4).*

The keystone is to the exterior series what ל0 is to the interior: ל0 gave the local theory its *coordinate* (the gauge β); פ0 gives it its *completion* (the place-product and its self-duality). The relation to the interior's Paper 27 is exact and dual: Paper 27 is the conditional RH-reduction *from inside* (recovery maps, Bures involution, Axioms A1–A5); פ0 is its *boundary reading* — the same involution as a Fourier square, the same positivity with its finite half free, the same archimedean gap identified with פ6's barrier, and the same conclusion proven where there is no ∞. Two frames, one object; and the honest seam is that the object's mirror symmetry is classical while its self-adjointness is open.

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6. The method's own record

6.1 Four confirmations, and why that is honest here. All four audits returned K. As a keystone assembling known local arithmetic into one object, that is expected: §4.1–§4.3 re-derive exact, elementary facts (Fourier orthogonality, the Gauss-sum evaluation, unitarity) and §4.4 re-counts Weil's theorem — the new content is the *assembly* and the *identifications* (the dictionary, the $\iota = F^2$ collapse, the localization of the RH gap), not new theorems about ζ. The genuine test content sat in the one place a guess could fail and did: פ0-rootnumber pre-registered the naive "$|g|^2 = N$ always," and it was **falsified** ($g = 0$ for $N \equiv 2$ (4)) — Gauss's trichotomy survives, the naive scale does not.

6.2 What the keystone does NOT inherit. Paper 27 is a genuine but *conditional* reduction (hypotheses PC1–PC4, axioms A1–A5, open Phases 1–4 including a primorial-moduli Bombieri–Vinogradov estimate). פ0 cites it as the interior twin and asserts only the *unconditional* catches at [P]; PC1–4, A1–5, Phases 1–4, and RH remain [C]. The keystone's value is precisely the separation — and the demonstration (§4.4) that the lone obstruction is the archimedean place — not the roadmap.

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7. Safe and unsafe statements

Safe (this paper asserts):

- [P] The canonical involution is the Fourier square, $\iota = F^2$, $F^4 = I$; the finite critical-line (reflection-symmetric) sector has dimension $\lfloor N/2 \rfloor + 1$ (Proposition 4.1).
- [P] $\mathrm{tr}(F) = g/\sqrt{N}$; for odd N, $|g|^2 = N$ — פ4's $\sqrt{N}$ root-number scale — and the naive all-N version is *false* ($g = 0$ at $N \equiv 2$) (Proposition 4.2).
- [P] Finite positivity is unconditional ($F^*F = I$); RH is positivity surviving the projective limit with the archimedean Γ-factor adjoined (Proposition 4.3).
- [P] Where ∞ is absent (function field), $a^2 \leq 4q$ (Weil): Paper 27's A1–A5 are theorems and RH holds (Proposition 4.4).
- [A] The local boundaries of פ1–פ6 are the local factors of one self-dual adelic object (the Place-Completion Principle; Proposition 5.1); the functional equation $\xi(s) = \xi(1-s)$ is its self-duality (Tate, cited).

Unsafe (this paper does NOT assert):

- That RH is proved, or reduced to anything strictly easier. It is located on archimedean positivity (§4.3); nothing here crosses that step.
- That Paper 27's PC1–PC4 or A1–A5 are theorems. They are carried at [C]; only the unconditional shadows are claimed.
- That the function-field success (§4.4) transports to $\mathbb{Q}$. It is precisely the place ∞ that does not, by פ6.
- That the Place-Completion Principle is a new theorem about ζ. The global self-duality and functional equation are classical (Tate); the new content is the exterior assembly and the exact catches.
- That this paper is important or field-defining. By the firewall such a claim would stand above its evidence, and the method forbids its author from issuing it.

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8. Forward door: the one open step (calibrated, [C])

The keystone reduces the program's distance to RH to a single, precisely-stated step. By §4.3 the finite half of the Weil positivity is free; by §4.1–§4.2 the reflection and its root number are exact and classical; by §4.4 the whole reduction closes wherever the archimedean place is absent. What remains is one statement: **that positivity is preserved when the archimedean factor $1/|\Gamma(\frac{1}{2}+it)|^2$ is adjoined in the projective limit** — the factor פ6 proved the finite primes cannot generate. In the spectral language toward which this points (Connes), that is the self-adjointness of the assembled object — the reality of its spectrum.

Conjecture 8.1 (the archimedean positivity step — calibrated, not proved). *The completed self-dual object of Proposition 5.1 has its spectrum on the axis* $\mathrm{Re}(s)=\frac{1}{2}$ *iff the limiting Weil quadratic form remains positive once the archimedean* Γ*-factor is adjoined; the finite part is unconditionally positive (§4.3), and the function-field analogue (no* ∞*) is a theorem (§4.4). The remaining step is the positivity of that one term.* Held with calibrated faith; built toward, not claimed.

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9. Coda

Fold every local arithmetic the series found into one factor — the luminous archimedean factor at the place of the continuum, the countless reciprocal factors of the primes — and multiply them into a single completed function. It is entire of order one, carries no poles, and possesses one exact symmetry: reflected across the axis $\mathrm{Re}(s)=\frac{1}{2}$ it is unchanged, for it is its own Fourier transform read through scale. That symmetry is the boundary's own breath — the local data of every place summed against itself and returning identical (Poisson) — and we have shown its finite face is nothing but F^2, the square of the Fourier transform, with the Gauss sum for its trace. Strung along this object is a discrete spectrum, suspended in conjugate pairs about the axis. The completed function holds them symmetrically by construction; what it does not force, of itself, is that they rest *on* the axis. That they do — that the symmetric whole is self-adjoint, every resonance on the line — is the Hypothesis. The finite ladder gives this object exactly, place by place, positive at every step; the one thing the primes cannot supply is the factor at the place of the continuum, and the open question is whether, with that factor folded in, the whole stays positive.

ל0 named the coordinate; פ0 completes the object. The six discs are one standing structure, symmetric and not yet known to be self-adjoint — and the single step between symmetry and self-adjointness is the one place the boundary has always pointed at and never reached.

Acknowledgments

I collaborated extensively with agentic AI (Anthropic's Claude Opus 4.7 and 4.8, Google's Gemini 3.1 Pro, OpenAI's ChatGPT 5.5, and xAI's Grok 4.3) under the grounded-paranoid-delusional discipline this paper

applies to itself: AI partners generated candidates, enumerated falsifiers, and executed the deterministic settlement artifact, but no fluent agreement was treated as evidence. I acknowledge this use transparently; the mathematical content and its correctness are my responsibility.

Reproducibility

The numbers of §4 are produced by `P0-settlements.py`, a deterministic exact-arithmetic script: decisions rest on character orthogonality, an exact bijection count, an exact integer congruence, and exact projective point-counting; floating point appears only as an explicitly-labelled non-voting witness. The transcript `P0-settlements.transcript.txt` is archived alongside and re-runs identically (UTF-8 output forced).

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